CHAPTER 14

PLANNING AND SUSTAINABLE DEVELOPMENT IN INDIAN CONTEXT

CHAPTER ESSENCE AT A GLANCE:
Examine sectoral versus target area planning, the Drought Prone Area Programme (DPAP), the socio-economic empowerment of the Gaddi tribe under the Integrated Tribal Development Project (ITDP) in Bharmaur, the Brundtland Commission report, and sustainable measures in the Indira Gandhi Canal Command Area.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Approaches to Planning:** Planning is either Sectoral (formulating schemes for economic sectors like agriculture, irrigation, industry, transport) or Target Area and Target Group Planning (tailored schemes for ecologically fragile or socio-economically marginalized regions to bridge regional disparities).

• **Target Area Programmes:** Command Area Development (CAD) and Drought Prone Area Programme (DPAP). DPAP was launched during the Fourth Plan to create employment and drought resilience through afforestation, pasture development, and watershed soil conservation.

• **Case Study: ITDP in Bharmaur Region:** Implemented for the transhumant Gaddi tribal community in Chamba district of Himachal Pradesh. Prioritized roads, potable water, electricity, schools, and hospitals; raised female literacy from 1.88 percent in 1971 to 65 percent in 2011.

• **Sustainable Development and Indira Gandhi Canal:** Defined by Brundtland Commission (1987) as meeting present needs without compromising future generations. Indira Gandhi Canal transformed Rajasthan desert, but over-irrigation triggered waterlogging and salinity, necessitating strict water management and shelterbelts.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

Target Area Approach $\Longleftrightarrow$ Command Area Development (CAD) + Drought Prone Area Programme (DPAP)
Regional Target Area Planning Framework

Integrated Tribal Development Project (ITDP) $\rightarrow$ Bharmaur Gaddi Tribal Region Eco-Preservation
Tribal Sub-Plan Socio-Economic Transformation

Brundtland Commission Definition (1987) $\Longleftrightarrow$ Development Meeting Present Needs Without Compromising Future
Foundational Principle of Sustainable Development

Indira Gandhi Canal (Stage I and II) $\Longleftrightarrow$ Harike Barrage (Punjab) $\rightarrow$ Thar Desert Irrigation Transformation
Arid Hydraulic Engineering Canal Lifeline

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Indira Gandhi Canal (formerly Rajasthan Canal) draws water from Harike Barrage located at the confluence of the Satluj and Beas rivers in Punjab, not from Bhakra Dam.

• **Alert:** The Gaddi community in Bharmaur practiced transhumance, migrating seasonally with livestock between alpine pastures in summer and lower Kangra valleys in winter.

• **Alert:** The Brundtland Report entitled 'Our Common Future' was published by the World Commission on Environment and Development (WCED) in 1987.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Differentiate between sectoral planning and target area planning with examples. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Sectoral Planning:** Entails formulation and execution of development schemes for distinct individual economic sectors such as agriculture, power, irrigation, or transport. **[1.5 Marks]**

• **Target Area Planning:** Focuses special development resources on specific backward, fragile, or tribal geographic regions (such as DPAP or Hill Area Development) to correct regional imbalances. **[1.5 Marks]**

2. Explain the main objectives of the Drought Prone Area Programme (DPAP) introduced in India. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Employment Generation:** Creating productive employment assets for drought-affected rural populations during the Fourth Five Year Plan. **[1 Mark]**

• **Drought Hardening:** Mitigating drought impacts through afforestation, pasture restoration, soil conservation, and watershed percolation tanks. **[1 Mark]**

• **Ecological Balance:** Restoring degraded ecological balance by integrating crop cultivation with livestock rearing and water conservation. **[1 Mark]**

3. Describe the socio-economic achievements of the Integrated Tribal Development Project (ITDP) in the Bharmaur region of Himachal Pradesh. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Geographic Setting:** Encompasses Bharmaur tehsil in Chamba district, the homeland of the semi-nomadic Gaddi tribe characterized by harsh winters. **[1 Mark]**

• **Infrastructure Development:** Prioritized vehicular road connectivity, piped drinking water, village electrification, and healthcare centers. **[1.5 Marks]**

• **Social Transformation:** Literacy surged dramatically; female literacy skyrocketed from a negligible 1.88 percent in 1971 to 65 percent in 2011, accompanied by a sharp decline in child marriage. **[1.5 Marks]**

4. Define sustainable development according to the Brundtland Commission Report 'Our Common Future' (1987). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** 'Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.' **[1.5 Marks]**

• **Inter-Generational Equity:** Emphasizes conservation of finite natural resources, ecological regeneration, and social equity between present and unborn generations. **[1.5 Marks]**

5. What environmental problems have been generated by intensive canal irrigation in the Indira Gandhi Canal Command Area? [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Severe Waterlogging:** Continuous excessive flood irrigation in an arid basin with impermeable subterranean clay layers caused the water table to rise rapidly to the root zone. **[2 Marks]**

• **Soil Salinization and Alkalization:** Capillary action drew dissolved salts to the desert surface, encrusting agricultural fields in sterile white saline deposits and degrading land fertility. **[2 Marks]**

6. Suggest any four ecological measures to promote sustainable development in the Indira Gandhi Canal Command Area. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **1. Strict Water Management:** Enforcing rotational warabandi water distribution and preventing wasteful flood irrigation. **[1 Mark]**

• **2. Drought-Resistant Cropping:** Shifting from water-intensive rice and wheat toward drought-hardy bajra, gram, and fodder crops. **[1 Mark]**

• **3. Afforestation and Shelterbelts:** Planting shelterbelts along canals and sand dunes to halt desert encroachment. **[1 Mark]**

• **4. Canal Lining:** Lining watercourses to arrest seepage and reclaim waterlogged agricultural tracts. **[1 Mark]**

7. Who are the Gaddis? In which physiographic region do they reside? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Identity:** A scheduled tribal pastoral community practicing transhumance with herds of sheep and goats, speaking the Gaddiali dialect. **[1 Mark]**

• **Region:** Reside in the Bharmaur tribal region of the Dhauladhar and Pir Panjal ranges along the Ravi river in Chamba district of Himachal Pradesh. **[1 Mark]**